Industry Workflows

We have selected two genuinely distinct industries, the same generic engine must support both without modification to any engine class and the workflows must differ in shape, participants, routing logic, failure modes and external systems.

Service Provisioning & Activation (NAICS 517311: Wired Telecommunications Carriers)

Organisation or Environment

A telecommunications carrier sells and delivers connectivity services (mobile subscriptions, enterprise circuits and fibre connections) to residential and small-business customers across multiple regions. The operator has network infrastructure only in serviceable regions, a field workforce for technician-assisted installations, and wholesale interconnections with credit bureaus, number/IP pool managers, an OSS/BSS platform, and an SMS gateway for customer notifications. In South Africa, the identity verification step must capture RICA-compliant evidence before the service can be activated. Order volume varies by marketing campaign, region, and season. Regulatory requirements (number portability, lawful intercept, KYC evidence retention) differ by jurisdiction, so the workflow must adapt to the region without modifying the engine.

Process Being Managed

When a customer places an order, the operator must first capture the order and verify the customer's identity to KYC/RICA standards. A credit assessment then determines whether the customer can be onboarded, referred for manual review, or declined. If acceptable, coverage is checked at the service address; if covered, network resources (a number or IP) are reserved. Depending on the order type, the service is either shipped for self-installation or a field technician is scheduled to install the CPE on site. Once the physical layer is ready, the network is configured, a billing account is opened, and the service is activated. The customer then confirms the service is working before the order is closed.

If the customer fails credit or has no coverage, the order is cancelled with a clear notification. If provisioning fails at the network layer, the workflow escalates to a network engineer and then a NOC manager. The workflow's central feature is the install-type fork: one branch ships physical hardware; the other dispatches a technician and requires an on-site installation step. Exactly one branch runs and the other is skipped.

Participants & Roles

Role	Responsibility
Sales Agent	Captures the order, verifies identity, coordinates with the customer, and asks the customer to confirm the activated service

Credit Analyst	Reviews credit cases that the automated assessment referred for manual decision
Customer Care	Schedules the field technician and acts as reviewer for the customer-confirmation step
Logistics	Ships the self-installation kit (CPE) to the customer
Field Technician (Gauteng)	Installs the CPE on site for orders in the Gauteng region
Field Technician (Western Cape)	Installs the CPE on site for orders in the Western Cape region
Network Engineer (L1 escalation)	Handles provisioning failures that need engineering attention
NOC Manager (L2 escalation)	Handles repeated failures or unresolved provisioning issues
System	Performs automated credit checks, coverage checks, resource reservation, network provisioning, billing setup, and service activation
Administrator (TIVIDY-level role)	Manages the workflow engine and monitors running instances (not a service participant, unchanged per scenario)
Auditor (TIVIDY-level role)	Inspects compliance evidence recorded in the workflow history (not a service participant, unchanged per scenario)

The two field technicians illustrate that assignment is region-aware: the RegionalTechnicianAssignment strategy will only choose a technician whose region attribute matches the order's region data field.

Major Pieces of Work

1. **Intake**: Capture the order (customer reference, service address, product, install type). Verify the customer's identity and capture KYC/RICA evidence. Run the credit assessment. If the credit bureau refers the case, route it to a credit analyst; if declined, notify the customer and close the order.
2. **Feasibility**: Check service coverage at the order's address. If no coverage exists, notify the customer and close the order. If coverage exists, reserve a network resource (MSISDN or IP) from the regional pool.
3. **Fulfilment**: Branch by install type. Self-install orders ship the CPE to the customer. Technician orders schedule a field technician in the matching region and record the on-site installation. Exactly one branch runs; the other is skipped by the routing decision.
4. **Activation**: Configure the network element for the reserved resource. Open a billing account. Activate the service once both are complete. Ask the customer to confirm the service is working.

Several pieces of work are themselves sub-processes. Intake contains the credit referral path as a conditional branch. Fulfilment is a true fork between self-install and technician branches. Activation contains parallel work for network provisioning and billing, both of which must complete before activation can proceed.

Important Decisions & Routing Points

- Credit check outcome: Approved, referred, or declined. Approved proceeds automatically. Referred activates the manual review step. Declined activates the notification and skips all later items.
- Manual credit review outcome: Approved or declined, with the same downstream effect as the automated outcomes.
- Install type: SELF_INSTALL activates the shipping branch and skips the technician path; TECHNICIAN activates the scheduling branch and skips shipping. This routing is data-driven, it branches on a value known at instance start, not on an outcome produced by a task.
- Service region: Determines which field technician can be assigned. RegionalTechnicianAssignment matches on data["region"] and only considers technicians in that region.
- Customer cancellation: An external signal (customer-cancelled) published via Tividy::signal() cancels the entire instance mid-process.

Approvals & Dependencies

- Identity verification depends on the order being captured.
- Credit check depends on identity verification (a customer cannot be credit-assessed without a verified identity).
- Fulfilment depends on credit being resolved (declined credit stops the workflow before fulfilment begins).
- The physical-layer step (shipping or installation) depends on the install-type routing gate completing.
- Activation depends on the physical-layer step completing (you cannot provision a service whose customer has no customer-provided equipment).
- Customer confirmation depends on activation.
- Customer confirmation is a reviewed item: it requires submit() by the sales agent and approve() by customer care.

Possible Failures or Exceptions

Situation	How to Resolve
Credit bureau unavailable	Adapter returns retryable failure; retry policy applies; if still failing, item escalates to network engineer, then NOC manager
Customer identity missing	RequiredDataRule on verify_identity throws a validation error; the item stays Available
Credit declined by bureau or	Workflow cancels the order and notifies the customer; downstream items

analyst	are skipped
Network OSS unavailable during provisioning	Retry policy applies; on exhaustion, item escalates to network engineer, then NOC manager
Region has no technician available	RegionalTechnicianAssignment returns empty; install_cpe stays Available; escalation notifies the supervisor
Customer not reachable for confirmation	customer_confirmation stays Under Review; deadline monitor fires and escalates
Customer cancels after activation	External signal customer-cancelled cancels the instance; deactivation of the service is a separate downstream process

External Systems

- Credit bureau: returns a credit score and a band (APPROVED / REFER / DECLINED).
- Network OSS: provisions the network element and activates the service.
- Notifier: sends customer notifications at key transitions.

Each external system is reached through an ExternalConnector adapter. For demonstration purposes, we support activities such as failure injection, so the retry and escalation paths (or other relevant feature) can be demonstrated on command.

Expected Outcome of Workflow

Every provisioning instance ends in one of:

- Completed (activated): service is live and the customer has confirmed it works.
- Completed (declined): credit rejected; downstream work was skipped; customer was notified.
- Cancelled: customer withdrew via the customer-cancelled signal.
- Failed: a mandatory step failed permanently and could not recover.

The audit log retains every action, provisioning attempt, credit outcome, and field technician assignment.

Transaction Dispute Resolution (NAICS 522110: Commercial Banking)

Organisation or Environment

A retail bank issues credit cards to consumers and handles disputes on their behalf. Disputes are governed by the card scheme's rules (Visa, Mastercard) and by local payment regulation, the National Payment System Act and the same card scheme arbitration rules. The bank coordinates with the merchant's acquiring bank, with the card scheme's arbitration platform, and with its own core banking ledger. Every action in a dispute must be auditable for compliance and legal review. Dispute volume varies seasonally and spikes after large sales events. Regulatory deadlines are strict, provisional credit must be issued within 10 business days and the merchant response window is fixed.

Process Being Managed

A cardholder disputes a charge on their credit card. The issuing bank must:

1. Validate the dispute against scheme rules and record the customer's stated reason.
2. Issue a provisional credit to the customer within the regulatory window, protecting the customer while the dispute is investigated.
3. File the chargeback with the card scheme, which routes it to the merchant's acquirer and then to the merchant.
4. Give the merchant a fixed window to respond with evidence supporting the transaction.
5. Analyse the evidence from both sides and issue a final decision.
6. Allow the losing party to appeal within a fixed window.
7. Apply the final decision to the ledger, either making the provisional credit permanent (customer wins) or reversing it and reinstating the charge (merchant wins).

If the merchant fails to respond within their window, the dispute is decided against them by default. The merchant and customer want opposite outcomes, and neither party controls the timing of the other's response.

Participants & Roles Involved

Role	Responsibility
Customer	Files the dispute, provides supporting evidence, may appeal
Dispute Agent	Validates eligibility, issues provisional credit, files the chargeback
Merchant Representative	Responds to the dispute with evidence, may appeal
Dispute Analyst	Reviews evidence from both sides and proposes a decision
Senior Analyst	Approves analyst decisions under the four-eyes principle; hears appeals
Compliance Officer	Handles second-level escalation for regulatory or high-value cases
System	Runs automatic tasks (chargeback filing, ledger settlement)
Administrator (TIVIDY-level role)	Manages the workflow engine and monitors running instances (not a dispute participant, unchanged per scenario)
Auditor (TIVIDY-level role)	Inspects compliance evidence recorded in the workflow history (not a dispute participant, unchanged per scenario)

Input Data

At instance start: transaction_id, amount, reason_code, card_ref, merchant_name. The dispute cannot be filed without these.

Major Pieces of Work

1. Intake: The customer lodges the dispute with their reason code and supporting evidence. The dispute agent validates it against scheme rules and records the eligibility outcome.
2. Provisional credit: The dispute agent issues the provisional credit (a regulatory deadline-driven step). Once posted, the chargeback is automatically filed with the card scheme.
3. Contest: The merchant's representative responds with counter-evidence. The analyst reviews both sides and proposes a decision. The losing party may appeal within a seven day window, and the appeal is heard by a senior analyst.
4. Settlement: The final decision is applied automatically to the ledger, either the provisional credit becomes permanent (customer wins) or it is reversed and the charge is reinstated (merchant wins).

Several pieces of work are themselves sub-processes. Contest contains the merchant response, the analyst decision, and both parallel appeal windows plus the appeal review. Settlement contains the final ledger operation, which is irreversible in the sense that a posted credit cannot be "unposted" from the customer's account without a compensating entry.

Important Decisions & Routing Points

- Eligibility outcome: The dispute is eligible, ineligible, or requires additional evidence. Ineligible routes to immediate cancellation with a customer notification.
- Merchant response outcome: Merchant submits evidence, accepts liability, or fails to respond within the deadline. Accepts liability or silence routes directly to settlement for the customer; silence is a timeout default judgment.
- Analyst decision: Customer wins or merchant wins.
- Merchant appeal: Appeal, accept decision, or let the deadline lapse (timeout = no appeal).
- Customer appeal: Same structure, inverse logic.
- Appeal review: Final decision that overrides the analyst's initial proposal.

Approvals & Dependencies

- Provisional credit must be issued before the merchant response window opens (you cannot file a chargeback that has no ledger reference).
- The merchant response window depends on the chargeback being filed with the card scheme.
- The analyst decision depends on the merchant response window being closed (merchant responded or the deadline lapsed).
- The merchant and customer appeal windows run in parallel, both depending on the analyst decision.
- The appeal review depends on both appeal windows closing.
- Settlement depends on both appeal windows and the appeal review.
- The analyst decision is a reviewed item: it requires submit() by the dispute analyst and approve() by the senior analyst (four-eyes principle).

Possible Failures or Exceptions

Situation	How to Resolve
Customer provides insufficient evidence	Analyst requests additional evidence; if not provided within the window, dispute is denied
Merchant fails to respond within deadline	Deadline monitor fires; merchant forfeits; default judgment for customer
Core banking system unavailable	Adapter returns retryable failure; retry policy applies; if still failing, item escalates to senior analyst
Card scheme platform unavailable	Same as above; escalation to senior analyst, then compliance officer
Customer disputes the final decision	Appeal path; the case reopens under the senior analyst
Duplicate dispute for the same transaction	Duplicate detection via transaction_id; second instance cancelled at intake
High-value dispute exceeds analyst authority	Escalates to senior analyst at intake and throughout
Regulatory hold on the dispute	External signal customer-withdrew cancels the instance

External Systems

- Core banking ledger: posts provisional credit, reverses or finalises the entry on settlement.
- Card scheme platform: files the chargeback with the scheme for arbitration.
- Notifier: sends email/SMS to the customer and merchant at key transitions.

Each external system is reached through an ExternalConnector adapter; the engine never depends on a vendor's protocol. (For demonstration purposes, we have allowed certain aspects such as failure injection to demonstrate the retry and escalation paths (or some other relevant feature) on command).

Expected Outcome of Workflow

Every dispute instance ends in one of:

- Completed (customer wins): provisional credit becomes permanent, funds transferred from merchant.
- Completed (merchant wins): provisional credit reversed, charge reinstated on the customer's account.
- Completed (ineligible): dispute rejected at intake with reason recorded.
- Completed (merchant silent): merchant forfeited by deadline; default judgment for customer.
- Cancelled: customer withdrew via the customer-withdrew signal.

- Failed: mandatory step failed permanently and could not recover.

The audit log records every action, actor, state change, deadline lapse, and escalation. The result is a complete regulatory compliance trail for each case.

Engine Classes Reused Without Modification

Both industries are executed by the same engine. No engine class depends on any industry package, and no engine class branches on the industry being executed.

The generic engine classes used without modification by both workflows are:

Concern	Generic Classes
Work item tree	WorkItem, TaskItem, StageItem, WorkItemSpec
Lifecycle	WorkItemState + 8 concrete states, expire()
Definition to an instance	WorkflowDefinition, WorkflowInstance, InstanceMemento
Construction	WorkflowBuilder, StandardWorkflowBuilder, WorkflowDirector
Routing	RoutingStrategy, DependencyRouting, OutcomeTableRouting, DataValueRouting
Assignment	AssignmentStrategy, RoleBasedAssignment, DataDrivenAssignment
Escalation	EscalationHandler, RoleEscalationHandler
Events	EventPublisher, EventSubject, EventObserver, WorkflowEvent
Audit	AuditLog, AuditObserver
Notifications	NotificationObserver, LoggingNotifier
Commands	Command, InstanceCommand + 7 concretes, MacroCommand, CommandDecorator, RetryingCommand, ActionHistory, CommandFactory
Memento	WorkItemMemento, InstanceMemento
Adapters	ExternalConnector, ConnectorRegistry
Façade	Tividy
Clock	Clock, SystemClock, ManualClock
Deadlines	DeadlineMonitor
Progress	WorkItemVisitor, ProgressVisitor
Iterator	WorkItemIterator, DepthFirstIterator
UI	Tui, Console, StdConsole, ScriptedConsole, ClientProfile

The following are added per industry, achieving genericity as no engine class has to change to support either engine:

- Banking: BankingPack, DisputeDirector, five industry-specific task classes, two vendor stubs, two adapters, one ClientProfile.
- Telecom: TelecomPack, ServiceActivationDirector, two industry-specific task classes, two vendor stubs, two adapters, one assignment strategy (RegionalTechnicianAssignment), one ClientProfile.